Escape rate, Lyapunov exponent and the uncertainty exponent of a latent-reasoning model

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https://github.com/euvel/eqr-transient-chaos

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Abstract

Recurrent reasoning models have fractal settle-time basins [1]. For the EqR Sudoku model we measure the two rates behind them: the maximal Lyapunov exponent λ_1 and the escape rate κ of the reasoning transient. We then test the transient-chaos relation $f(\varepsilon) \propto \varepsilon^{\kappa/\lambda_1}$ for the fraction of ε-close starting latents that settle at different loops. $\lambda_1 = 2.07$ per loop is a property of the model (19 puzzles, uncorrelated with κ), whereas κ varies 20-fold across puzzles. At true float64, with ε from 10^{-1} to 10^{-12}, the measured exponent tracks κ/λ_1 on 9 puzzles (rank correlation 0.92; 8 of 9 within -23% to $+19\%$), including 4 puzzles used only after a pre-registered first test. Survival is exponential after a start-up of a few loops, and $f(\varepsilon)$ shows no steepening over 11 decades. For this model the transient is therefore ordinary transient chaos with non-slim boundaries, not doubly transient chaos.

1 Background

For an open chaotic system with escape rate κ and maximal Lyapunov exponent $\lambda_{\max}$, the uncertainty exponent α is defined by the fraction $f(\varepsilon) \sim \varepsilon^{\alpha}$ of initial conditions whose outcome changes under a perturbation of size ε. It obeys $\alpha = \kappa/\lambda_{\max}$, and the boundary of a 2D slice has dimension $2 - \alpha$ [2–4]. Heuristically, a perturbation ε grows to $O(1)$ after $t_\varepsilon \approx \ln(1/\varepsilon)/\lambda_1$ loops, and the outcome changes only if the trajectory is still in the transient by then: $f \approx S(t_\varepsilon) \approx e^{-\kappa t_\varepsilon} = \varepsilon^{\kappa/\lambda_1}$.

2 Setup

EqR Sudoku (checkpoint `locuslab/EqR-model`) via loopscape [1] at commit `16b52aa`, noise off, using the demo slice: a random orthonormal 2-plane in the 97×512 latent z_H, with $z_L = 0$. The settle time T is the last loop at which the decoded grid changes. κ is an exponential fit to $S(k) = P(T > k)$ on a 40×40 map with cap 150, kept if $R^2 \geq 0.97$. λ_1 is the median over the 8 slowest trajectories of Benettin twins with QR ($K = 4$, offset 10^{-4}). ε-pairs: a base point z_0 uniform on the slice and a partner $z_0 + \varepsilon\|z_0\|w$, with w a random unit direction in the full latent, 48 pairs per level. These runs use true float64. loopscape's float64 option silently runs in float32, and EqR's `rms_norm` casts to float32; both were patched and verified in-kernel (reported as loopscape#1).

3 Results

Rates. Over 19 puzzles λ_1 has median 2.07 per loop (IQR 1.93–2.25), with no correlation with κ (Spearman -0.01) or with the median settle time (-0.05). κ ranges from 0.049 to 1.08 per loop and is unrelated to the human difficulty rating (-0.14). Since λ_1 is fixed by the model, a puzzle's fractality $2 - \kappa/\lambda_1$ is set by its escape rate alone.

Exponent. Fig. 2 compares the slope of $\log f$ against $\log \varepsilon$ with κ/λ_1 measured independently. Puzzle by puzzle the ratios are 1.17, 1.04, 1.16, 0.83, 1.19 (v5) and 0.77, 0.84, 0.90, 0.48 (v6). The outlier, 149 ($\kappa/\lambda_1 = 0.70$), rests on only 2 usable levels. Both runs registered this comparison as a descriptive outcome without a pass bar.

Amplitude. The horizon law $f \approx \langle S(\ln(0.1/d_0)/\lambda_1)\rangle$, where d_0 is the measured separation after one loop, was the pre-registered prediction. In v5 it *failed* its bar (61% of points within $2\times$, bar 70%). Pooled over v5 and v6, 77% of 31 points fall within $2\times$, and 87% when each run's own survival curve is used. The remaining bias is $\approx -20\%$, consistent with pairs that decorrelate yet settle on the same loop. A post-hoc constant delay fitted on v5 was rejected out of sample in v6.

Not doubly transient, not slim. Lai et al. expect doubly transient chaos [1], in which the settling rate grows in time and the boundaries become slim [5,6]. In discrete-time hazard fits on 13 maps, every puzzle shows a start-up of 2–6 loops followed by a constant exit rate. All 11 late-window fits are exponential, and the pooled rate changes by $\times 1.11$ per 50 loops (95% upper bound $\times 1.36$); puzzle 1216 holds 0.048 per loop from loop 8 to loop 93. The local slopes of $f(\varepsilon)$ do not steepen: curvature $+0.006$ [-0.09, $+0.07$], and the largest small-ε slope is 0.22.

4 Limitations and open question

These results cover one model, one task and one slice, with 9 puzzles for the exponent. Slopes come from 2–5 levels. Survival is exponential within loop caps of 120–150. The relation $\alpha = \kappa/\lambda_{\max}$ does not distinguish *rough* from *filamentary* boundaries [4]: that requires the cross-boundary Lyapunov exponent, which we did not measure and which is the natural next experiment. Data, kernels, pre-registrations and a script for every number are in the repository.

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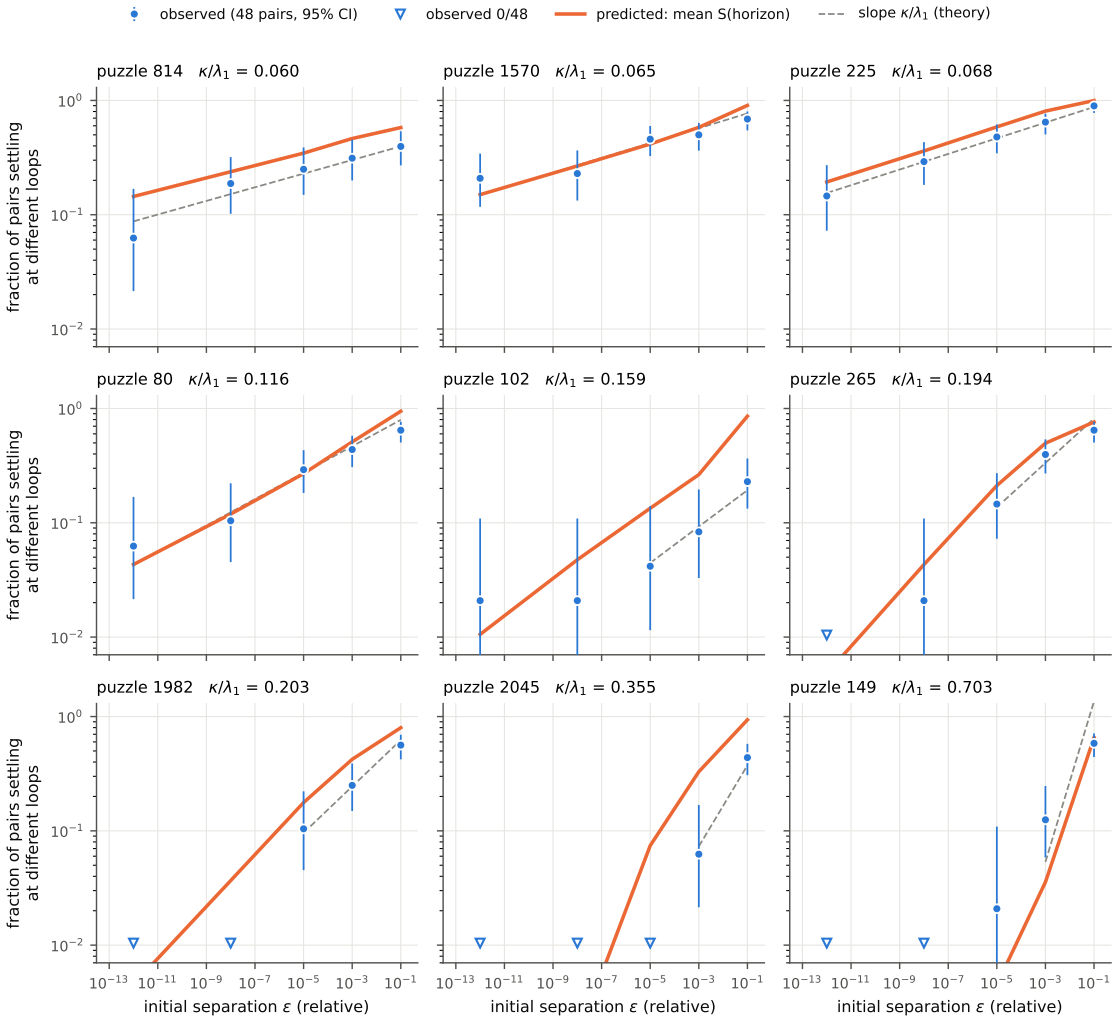


Figure 1: Fraction of 48 ε-pairs settling at different loops (blue; 95% Wilson intervals; triangles mark 0/48), the horizon prediction (orange), and a reference slope κ/λ_1 (grey). Panels are ordered by κ/λ_1.

responsibility for them, rest solely with the author.

References

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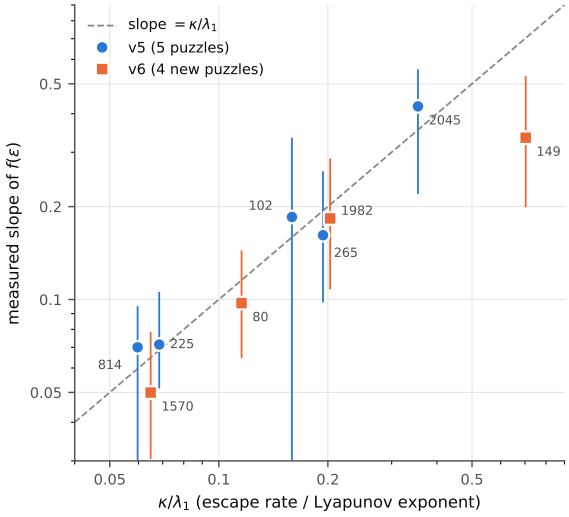


Figure 2: Measured exponent of $f(\varepsilon)$ against κ/λ_1 (95% bootstrap intervals over pairs). The dashed line is $\alpha = \kappa/\lambda_1$. The v6 puzzles were not used in v5.